End of chapter Questions: Pgs 136 - 137
Q: 1-4, 6-15

GLOBAL EDITION



MICROECONOMICS

non-necessity products have a higher-rate of elasticity

while necessity products

14th Edition

① Price Elasticity of Demand
② Income Elasticity

Michael Parkin

varies in denominator but same calculation of Demand apply



4 Elasticity

how much ppl are responding measures responsed the public as price changes

while everything remains constant

Price ↑ Qd ↓

e.g. demand for insulin is maintain

Qd for a necessity never decreases despite rises in the price

After studying this chapter, you will be able to:

- Define, calculate, and explain the factors that influence the price elasticity of demand
- Define, calculate, and explain the factors that influence the income elasticity of demand and the cross elasticity of demand
- Define, calculate, and explain the factors that influence the elasticity of supply

In luxury, Qd decreases as the price will increase
→ can be avoided / postponed

★ we must analyze the type of good by

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• big public response (stretch) = elastic

evaluating the demand

Price Elasticity of Demand

• small public response (unstretchable) = inelastic

You know that when supply decreases, the equilibrium price rises and the equilibrium quantity decreases.

But does the price rise by a large amount and the quantity decrease by a little?

Or does the price barely rise and the quantity decrease by a large amount?

The answer depends on the responsiveness of the quantity demanded of a good to a change in its price.

we will analyze:
- income
- cross prices (between substitutes & complementary items)
- prices

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- Elasticity is a unit-free measure
- Elasticity are never in negative
e.g -0.5 is written as 0.5

Price Elasticity of Demand

You might think about the responsiveness of the quantity demanded of a good to a change in its price in terms of the *slope of the demand curve*.

If the demand curve is steep, the price rises by a lot; if the demand curve is almost flat, the price barely rises.

But the slope of a demand curve depends on the units in which we measure the price and the quantity. We can choose these units to make the demand curve steep or flat.

To measure responsiveness we need a measure that is *independent* of units of measurement.

Elasticity is such a measure.

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Price Elasticity of Demand

The **price elasticity of demand** is a units-free measure of the responsiveness of the quantity demanded of a good to a change in its price when all other influences on buying plans remain the same. → while other shifters / factors remain constant

Calculating Price Elasticity of Demand

The price elasticity of demand is calculated by using the formula:

numerator = Qd ΔQd

$$\frac{\text{Percentage change in the quantity demanded}}{\text{Percentage change in the price}}$$

ΔPrice

denominator = Price

1) Calculate Percentage 2) Find price elasticity of demand 3) Interpret your

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(they are NEVER given)

$\frac{10\%}{5\%} = 2$

answer

Price Elasticity of Demand

- $\varepsilon > 1$ = elastic
- $\varepsilon < 1$ = Inelastic
- $\varepsilon = 1$ → unit elastic (both are changing by the same %)

e.g. Qd & Price both change by 10%

To calculate the price elasticity of demand:

We express the change in price as a percentage of the *average price*—the average of the initial and new price, …

and we express the change in the quantity demanded as a percentage of the *average quantity* demanded—the average of the initial and new quantity.

- COMMON MISTAKE: switching Qd / price from their positions in the fractions

$$\%\,\Delta d = \frac{\text{New} - \text{Old}}{\text{Midpoint}} \times 100$$

(Midpoint ↓)

$$\%\,\Delta \text{price} = \frac{\text{New} - \text{Old}}{\text{Midpoint}}$$

(Midpoint ↓ means a

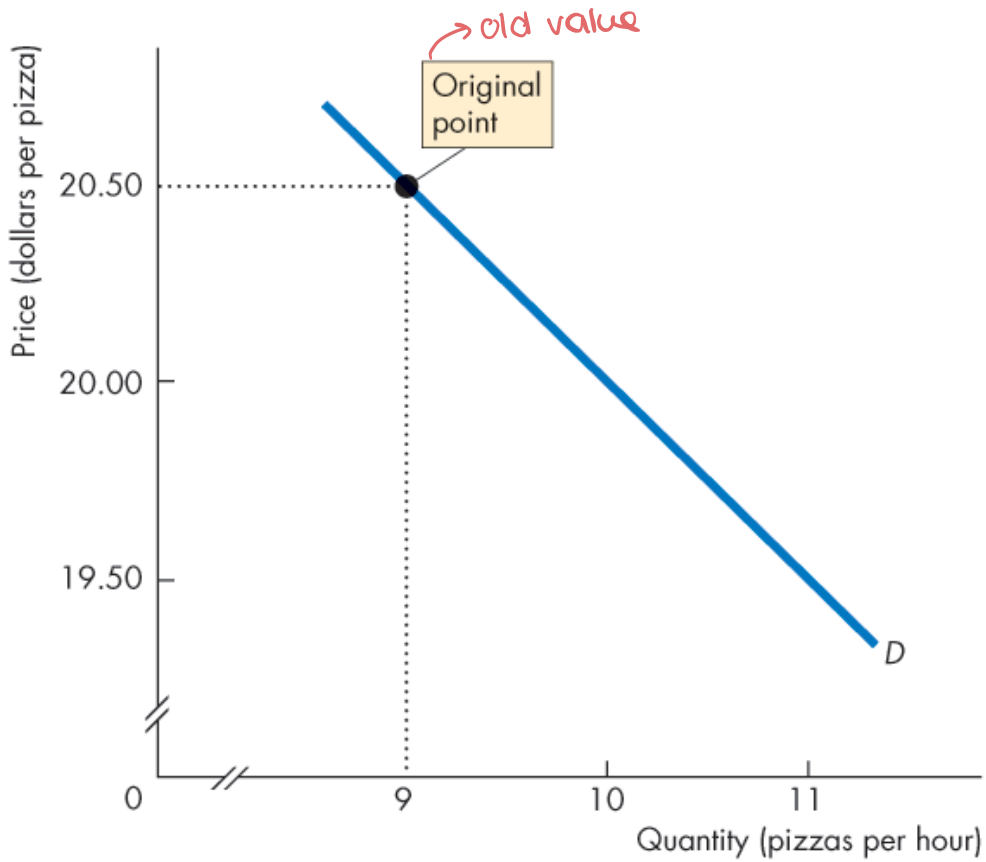
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means average

Price Elasticity of Demand

Figure 4.1 calculates the price elasticity of demand for pizza.

Initially, the price of a pizza is $20.50 and the quantity demanded is 9 pizzas an hour.



★ ALWAYS SOLVE YOUR ANSWERS STEP-BY-STEP

- schedule/ graph can be given to determine the percentage

. P = 70 , Qd = 500 (old)

. P = 90 , Qd = 300 (new)

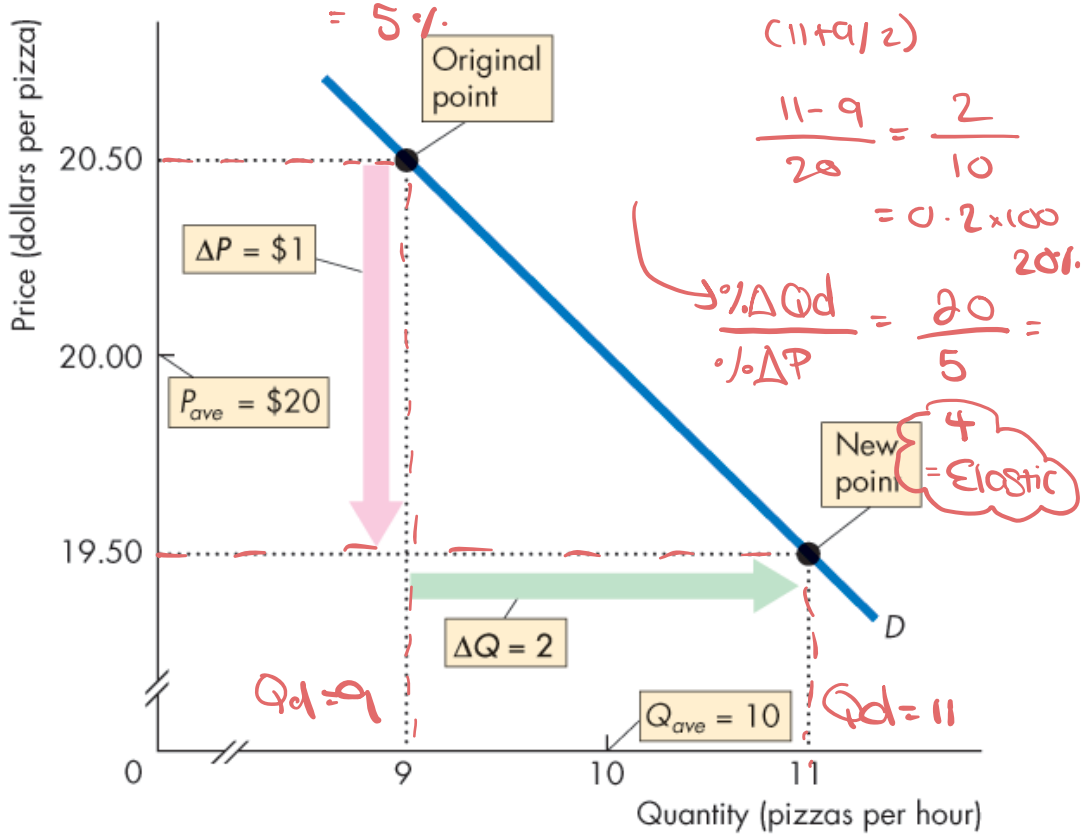
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Price Elasticity of Demand

The price of a pizza falls to $19.50 and the quantity demanded increases to 11 pizzas an hour.

The price falls by $1 and the average price is $20.

The quantity demanded increases by 2 pizzas and the average quantity is 10 pizzas.



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% Δ in Qd < % Δ in Price = Inelastic

% Δ in Qd = % Δ in Price = unit elastic

Price Elasticity of Demand

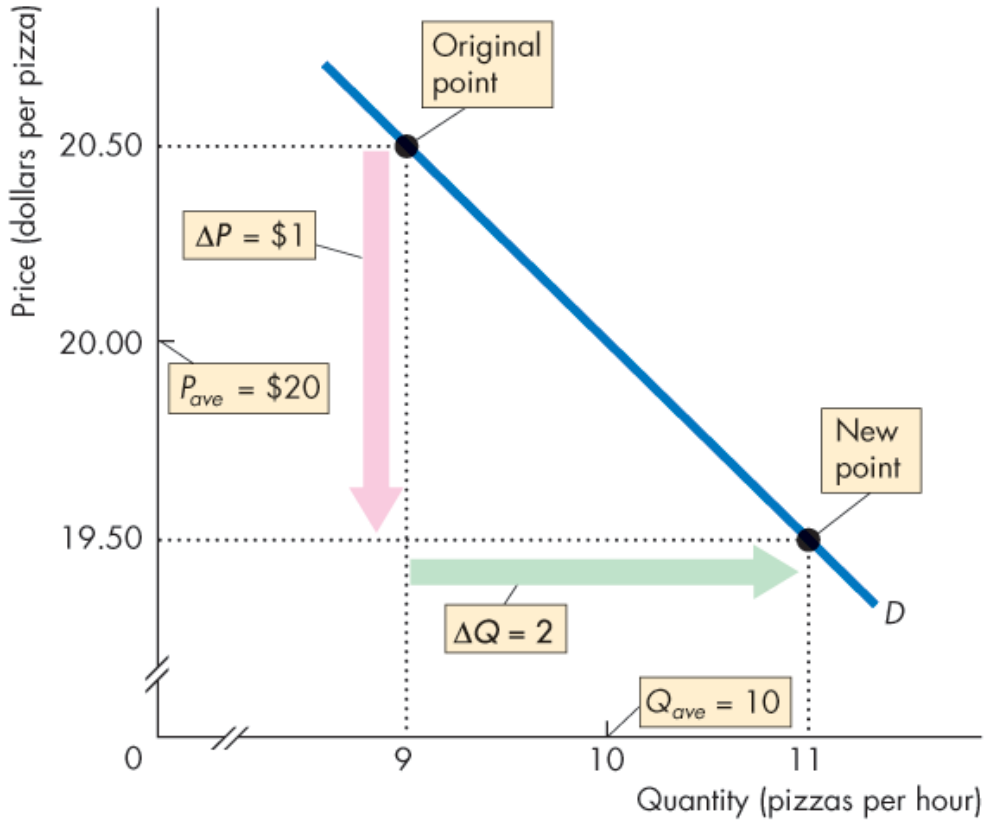
The percentage change in quantity demanded is $\Delta Q/Q_{ave}$ x 100, which is

ΔQ = New quantity - initial quantity

(2/10) x 100 = 20%.

The percentage change in price is $\Delta P/P_{ave}$ x 100, which is

($1/$20) x 100 = 5%.



Interpret based on Price

If Price changes by one unit, how will it impact the Qd

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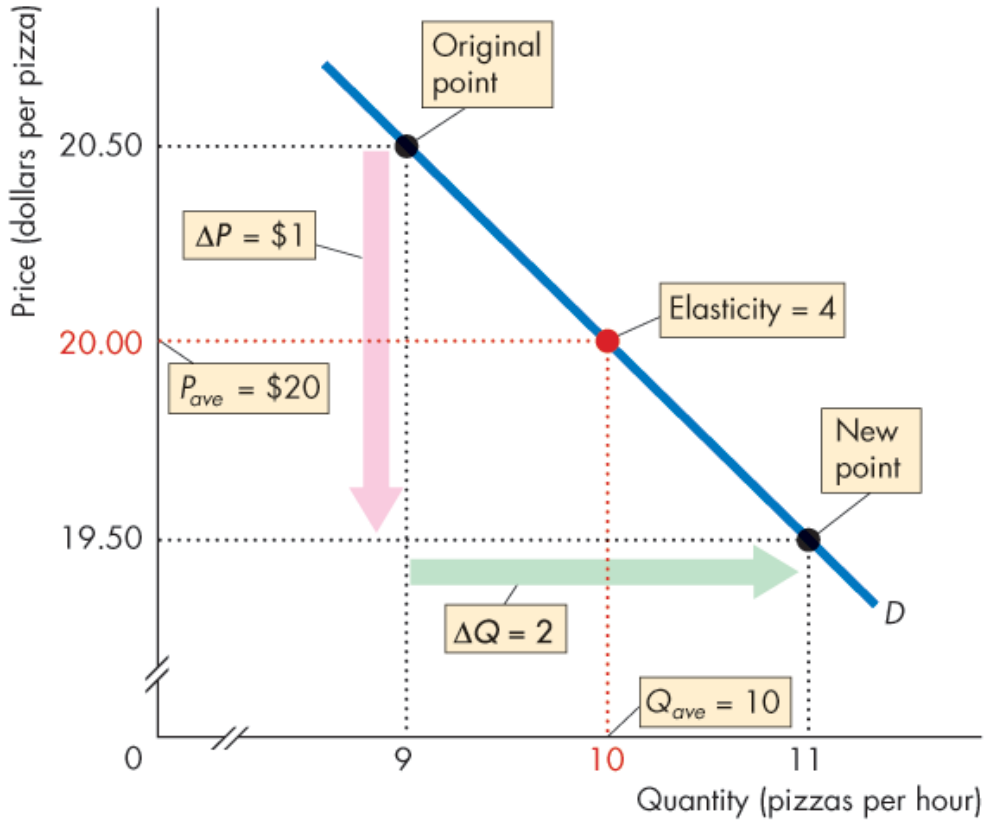
Price Elasticity of Demand

The price elasticity of demand equals:

The percentage change in quantity demanded divided by the percentage change in the price.

Price elasticity of demand equals 20% ÷ 5% = 4.

The elasticity equals 4 at the mid-point between the two prices.



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Price Elasticity of Demand

Average Price and Quantity

By using the *average price* and *average quantity*, we get the same elasticity value regardless of whether the price rises or falls.

A Units-Free Measure

Elasticity is a ratio of percentages, so a change in the units of measurement of price or quantity leaves the elasticity value the same.

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Price Elasticity of Demand

Minus Sign and Price Elasticity of Demand

The formula yields a negative value, because price and quantity move in opposite directions.

But it is the *magnitude*, or absolute value, that reveals how responsive the quantity change has been to a price change.

So for price elasticity of demand we use the absolute value and drop the minus sign.

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Price Elasticity of Demand

Inelastic and Elastic Demand

Demand can be inelastic, unit elastic, or elastic, and can range from zero to infinity.

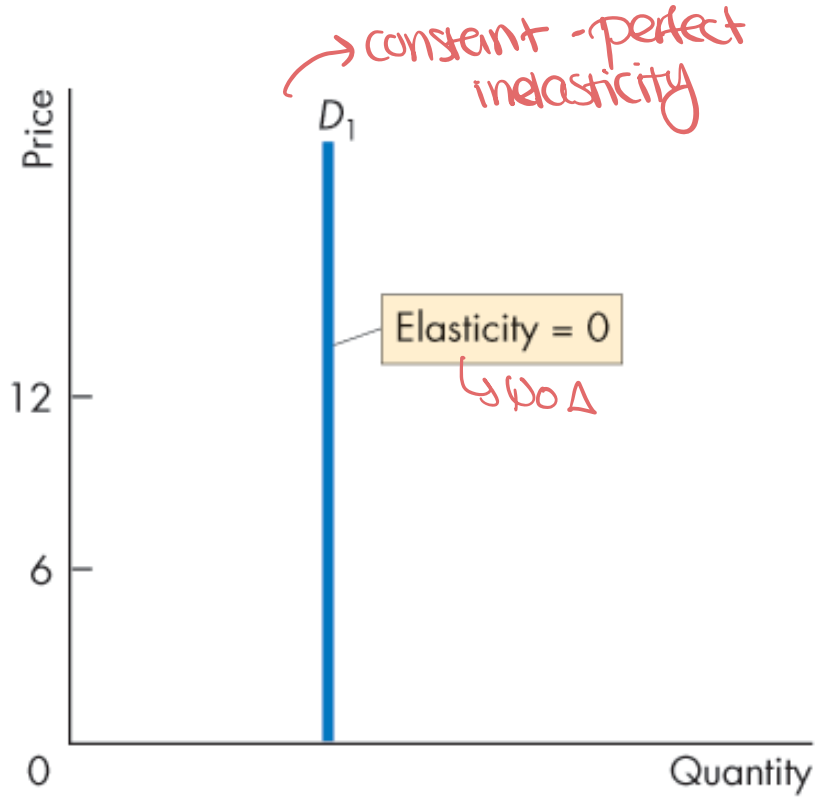
If the quantity demanded doesn't change when the price changes, the price elasticity of demand is zero and the good has a **perfectly inelastic demand**.

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Price Elasticity of Demand

Figure 4.2(a) illustrates the case of a good that has a perfectly inelastic demand.

The demand curve is vertical.



(a) Perfectly inelastic demand

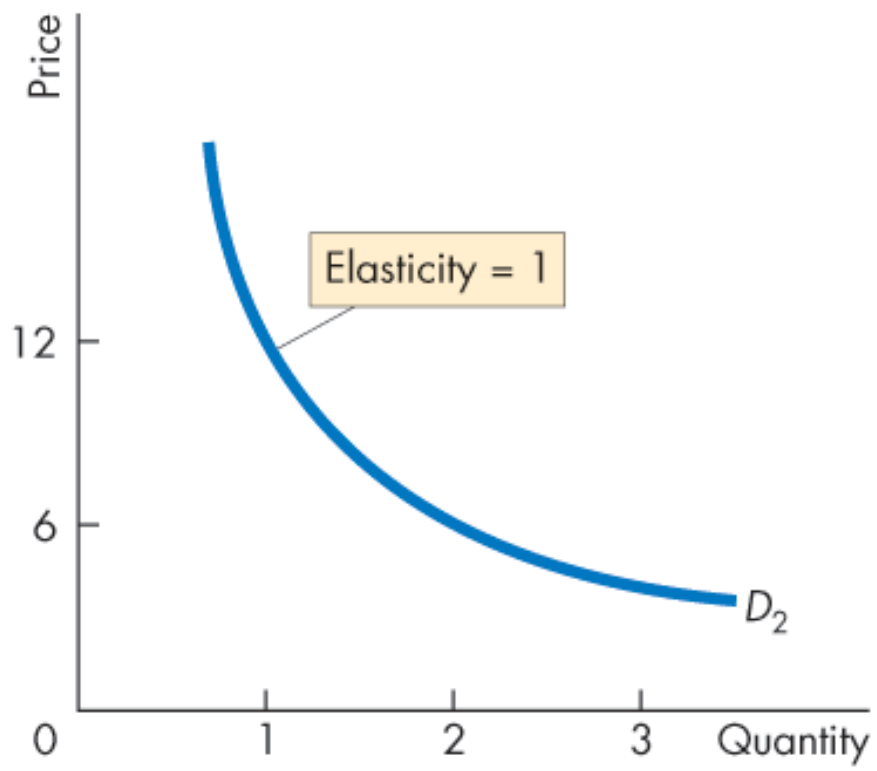
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Price Elasticity of Demand

If the percentage change in the quantity demanded equals the percentage change in price, …

the price elasticity of demand equals 1 and the good has **unit elastic demand**.

Figure 4.2(b) illustrates this case—a demand curve with ever declining slope.



(b) Unit elastic demand

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Price Elasticity of Demand

If the percentage change in the quantity demanded is *smaller* than the percentage change in price,

the price elasticity of demand is *less* than 1 and the good has **inelastic demand**.

If the percentage change in the quantity demanded is *greater* than the percentage change in price,

the price elasticity of demand is *greater* than 1 and the good has **elastic demand**.

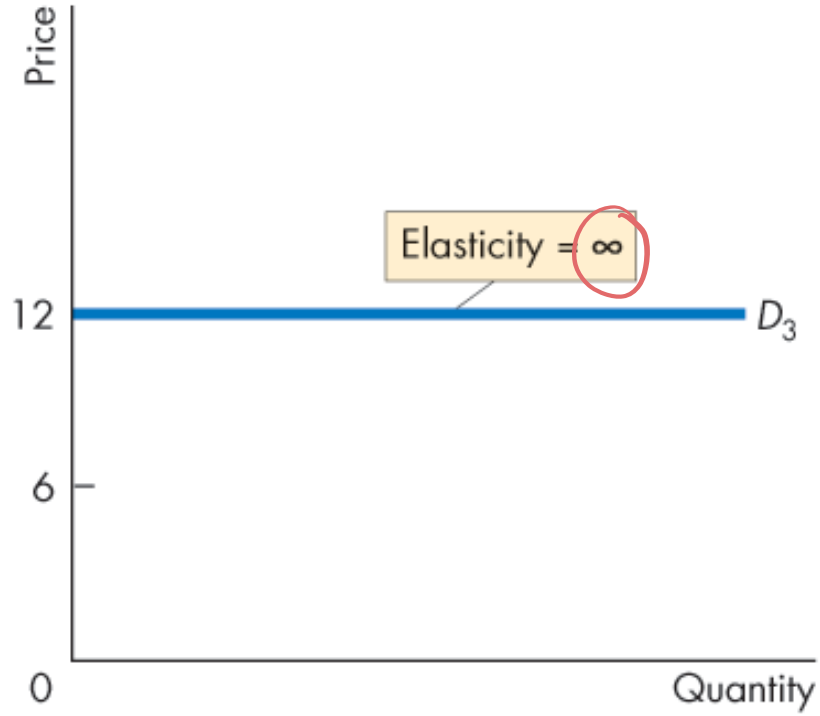
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Price Elasticity of Demand

If the percentage change in the quantity demanded is infinitely large when the price barely changes, …

the price elasticity of demand is infinite and the good has a **perfectly elastic demand**.

Figure 4.2(c) illustrates the case of perfectly elastic demand—a horizontal demand curve.

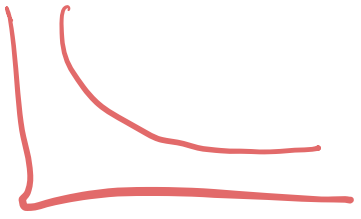


(c) Perfectly elastic demand

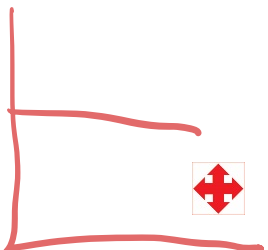
1) Inelastic



2) unit elastic



3) Elastic



Determinants influencing Elasticity

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- close substitutes → elastic
- income level: ↑ Income → elastic E.g. diabetes patients buying insulin

Price Elasticity of Demand

(Qd never changes with necessity goods)

- Time: short-run (10mins) & long-run (1 month) → enough time to make adjustments → Elastic

The Factors That Influence the Elasticity of Demand

Inelastic

The elasticity of demand for a good depends on:

The closeness of substitutes

The proportion of income spent on the good

The time elapsed since a price change

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Price Elasticity of Demand

Closeness of Substitutes

The closer the substitutes for a good or service, the more elastic is the demand for the good or service.

Necessities, such as food or housing, generally have inelastic demand.

Luxuries, such as exotic vacations, generally have elastic demand.

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Price Elasticity of Demand

Proportion of Income Spent on the Good

The greater the proportion of income consumers spend on a good, the larger is the elasticity of demand for that good.

Time Elapsed Since Price Change

The more time consumers have to adjust to a price change, or the longer that a good can be stored without losing its value, the more elastic is the demand for that good.

↓ storage life of a good inc customer ability to adjust their purchasing plans relative to price changes

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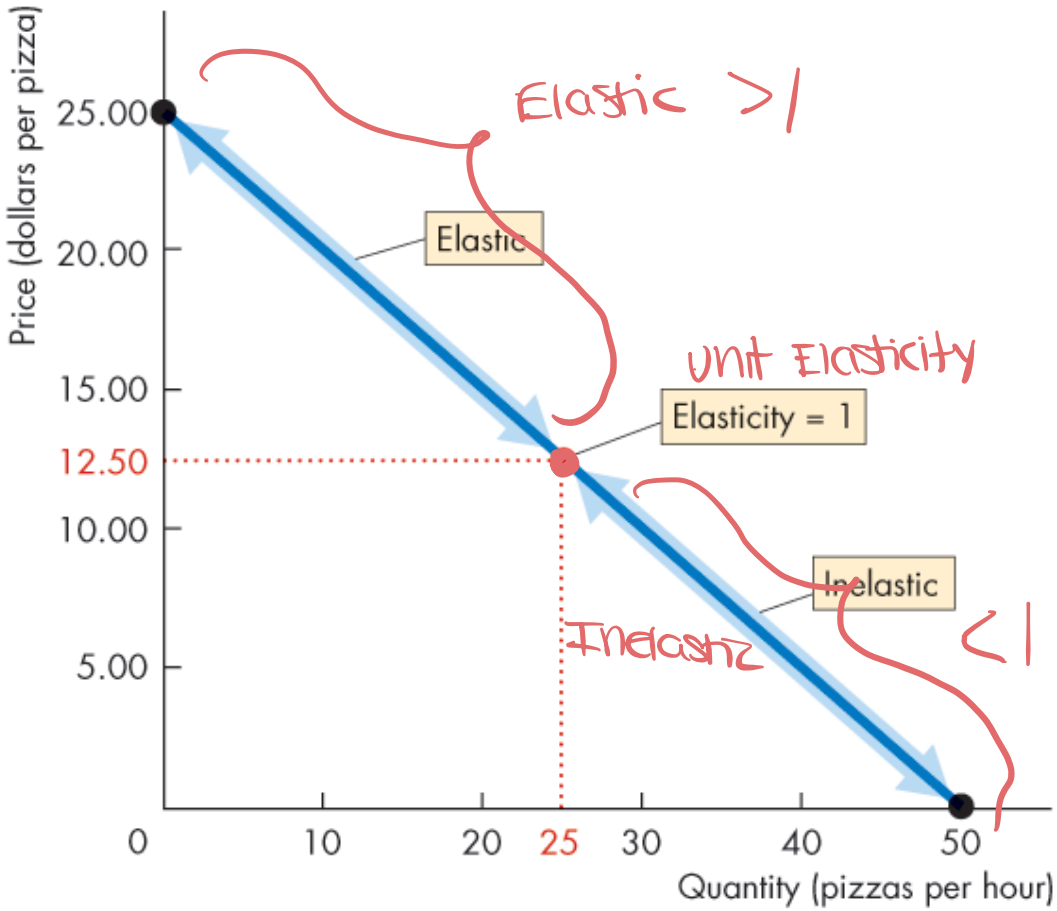
Price Elasticity of Demand

Elasticity Along a Linear Demand Curve

Figure 4.3 shows that at the mid-point of a linear demand curve, demand is unit elastic.

At prices above the mid-point, demand is elastic.

At prices below the mid-point, demand is inelastic.

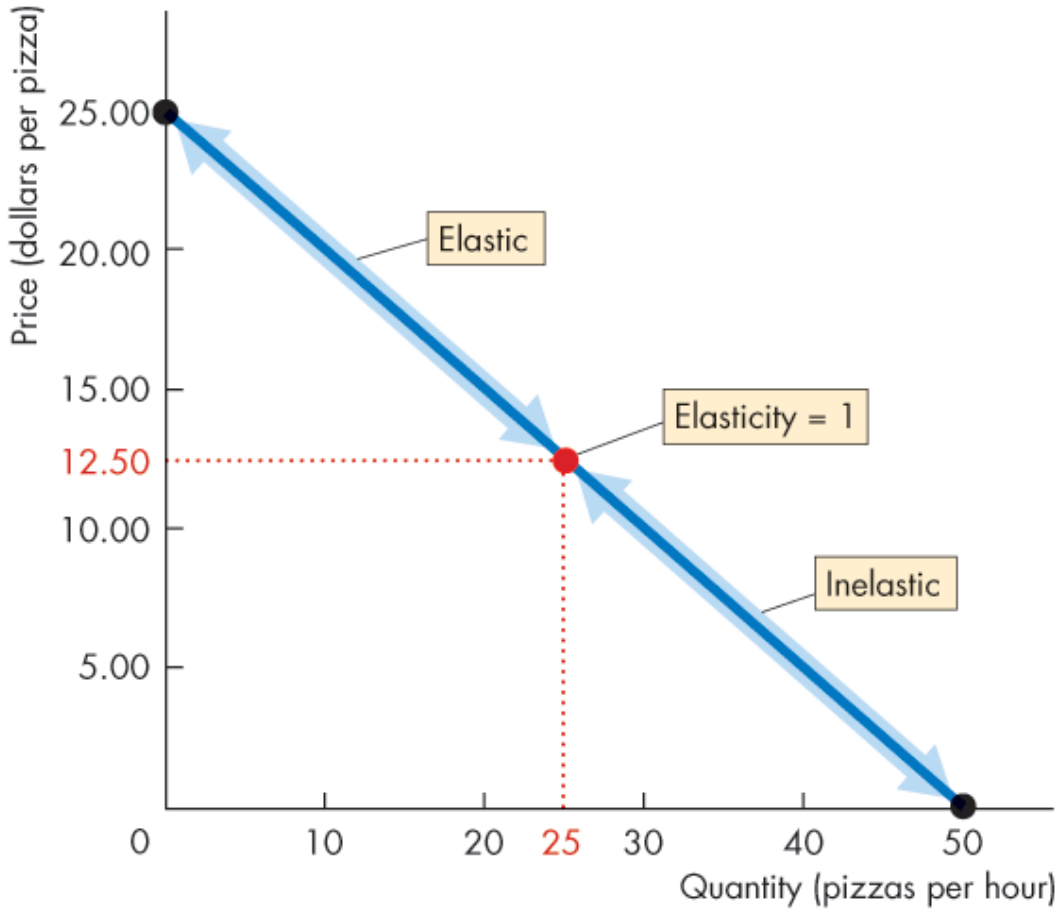


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Price Elasticity of Demand

Let's calculate the price elasticity of demand at different prices …

to confirm that the demand for pizza becomes more inelastic as pizza becomes cheaper.



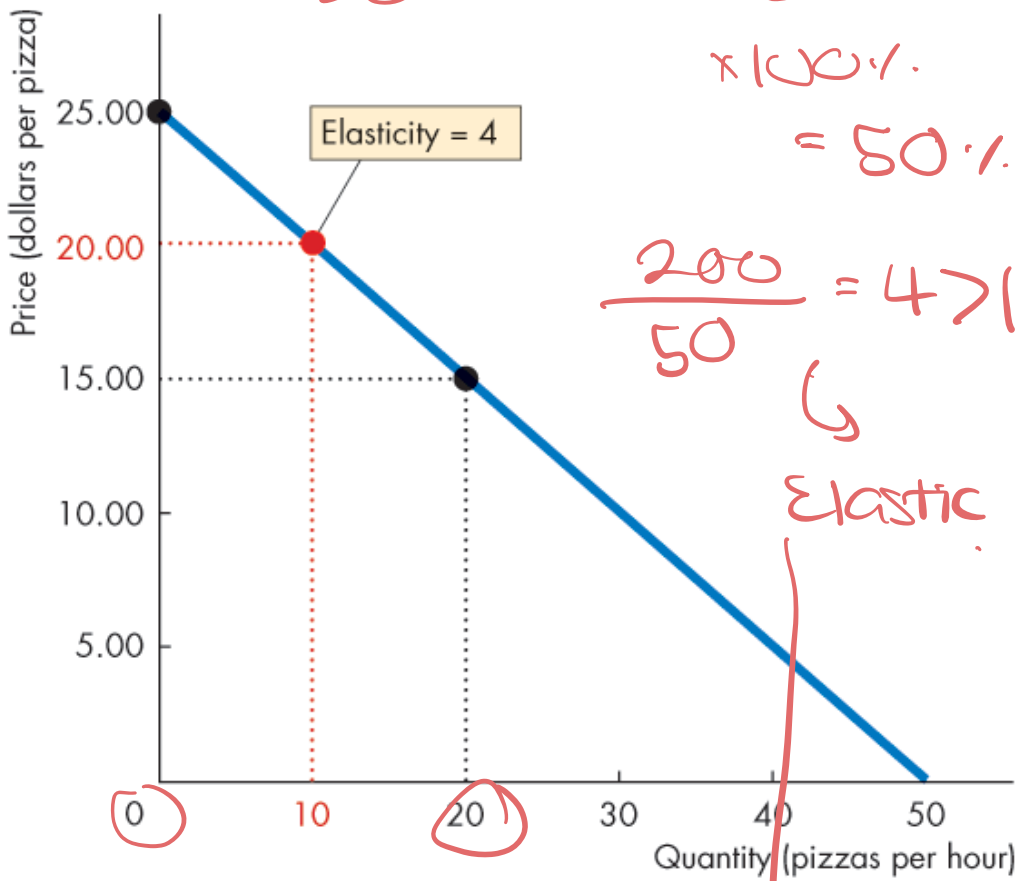
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Price Elasticity of Demand

If the price falls from $25 to $15, the quantity demanded increases from 0 to 20 pizzas an hour.

The average price is $20 and the average quantity is 10 pizzas.

The elasticity of demand is (20/10) ÷ (10/20), which equals 4.



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by 40% (or 4 times)

Price Elasticity of Demand

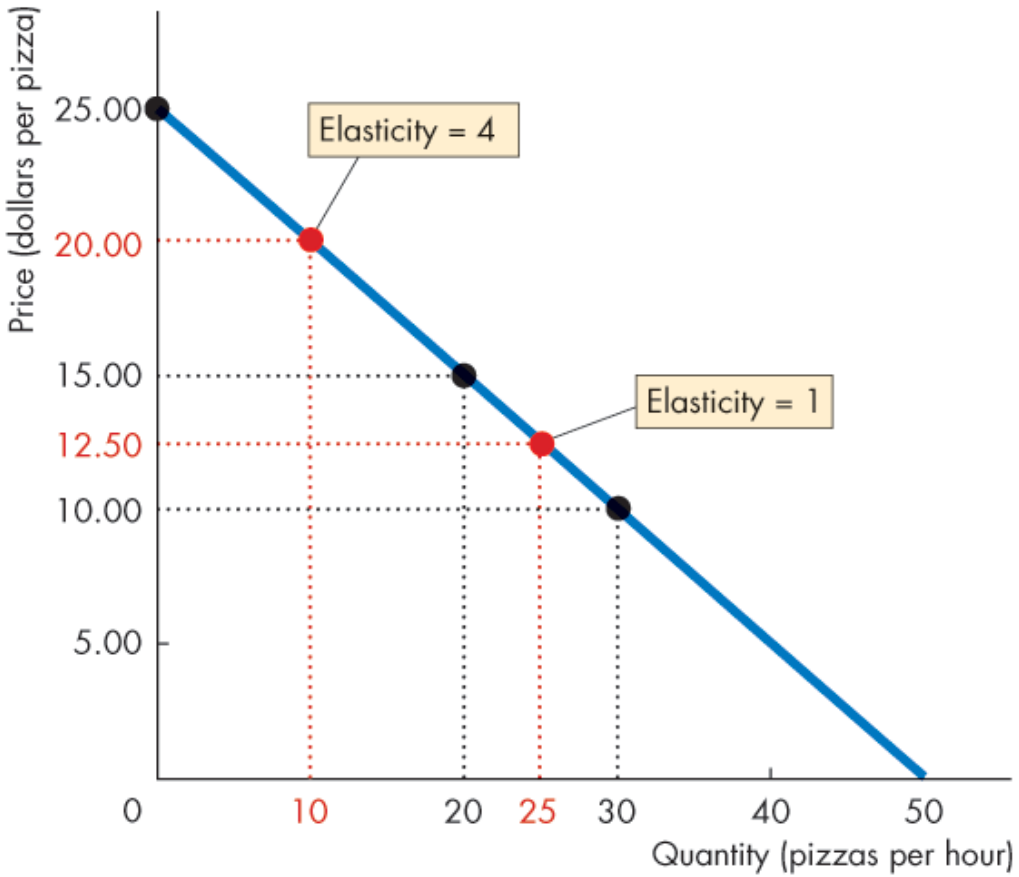
If the price falls from $15 to $10, the quantity demanded increases from 20 to 30 pizzas an hour.

old new

old new

The average price is $12.50 and the average quantity is 25 pizzas.

The elasticity of demand is (10/25) ÷ (5/12.50), which equals 1.



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- $\frac{\Delta Qd}{Qdave} = \frac{New - old}{Midpoint} = \frac{30-20}{25} = \frac{10}{25} = 0.4$ $\times 100$ $= 40\%$

  $30+20 = 50$, $\frac{50}{2} = 25$

- $\frac{\Delta Price}{Price_{ave}} = \frac{New - old}{Midpoint} = \frac{|10-15|}{12.5} = \frac{5}{12.5} = 0.4 \times 100 = 40\%$

  $10+15=25$, $25/2 = 12.5$

- $\frac{\%\Delta Qd}{\%\Delta Price} = \frac{40\%}{40\%} = 1$

  Unit Elastic

  for each unit price increase, the Qd increases by one unit respectively

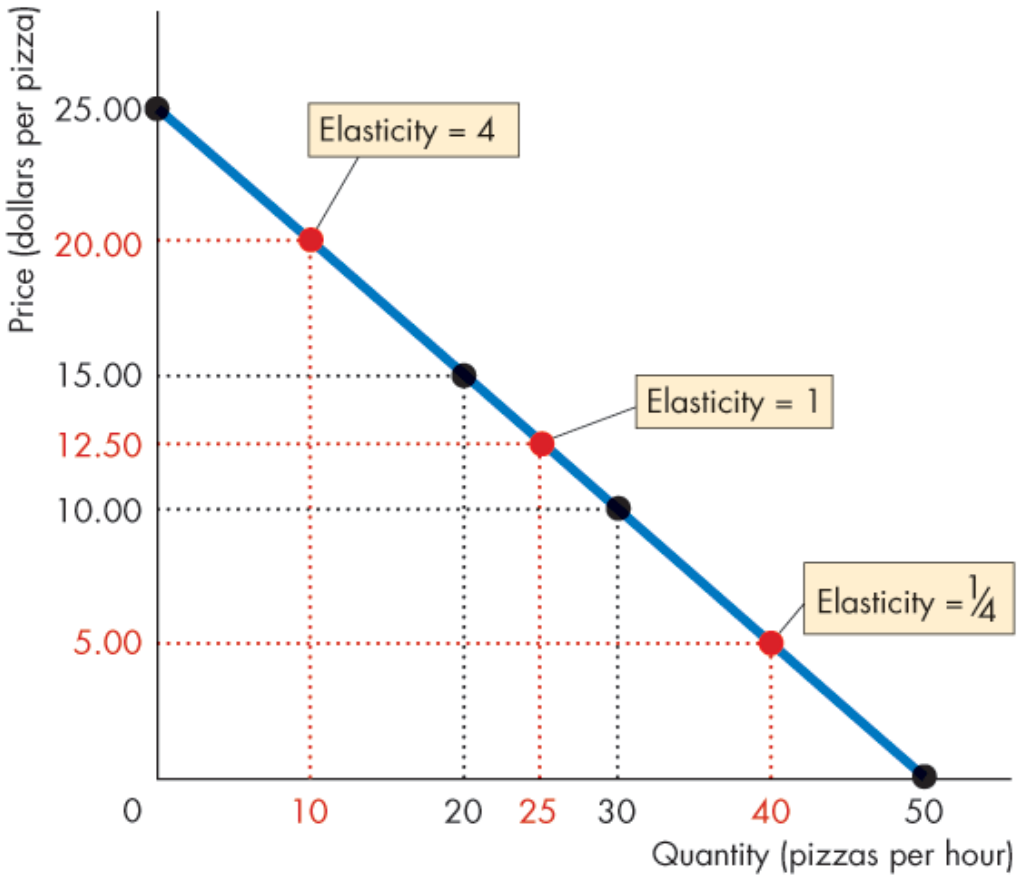
Price Elasticity of Demand

↑ HW

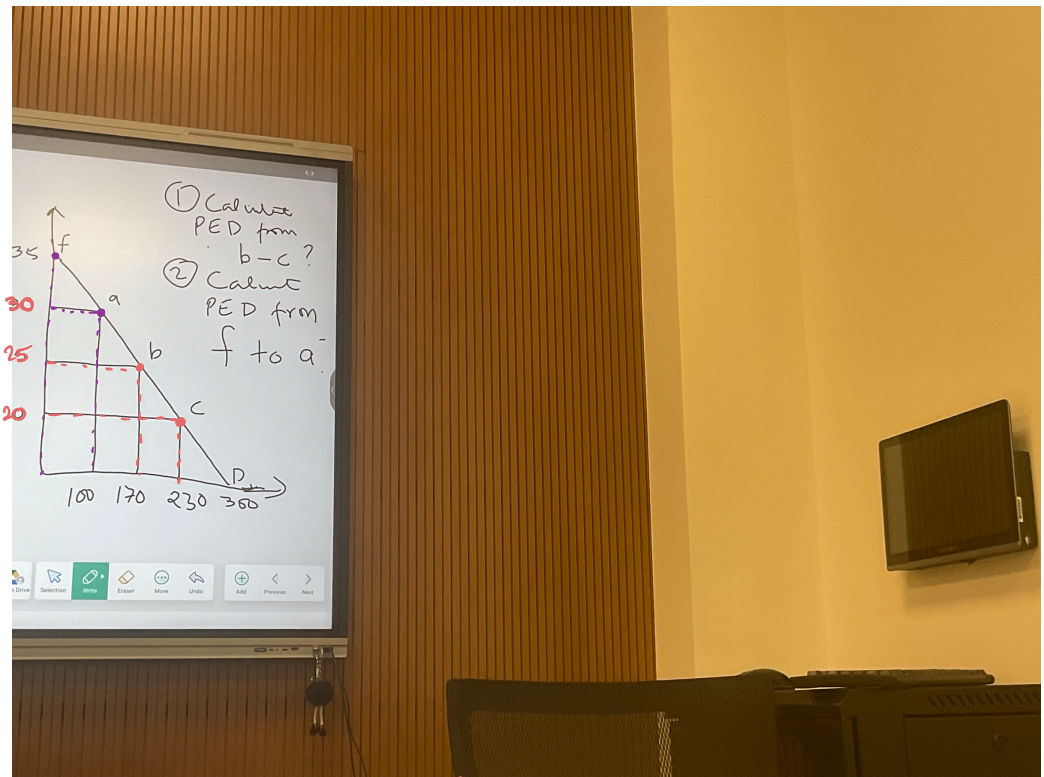
If the price falls from $10 to $0, the quantity demanded increases from 30 to 50 pizzas an hour.

The average price is $5.00 and the average quantity is 40 pizzas.

The elasticity of demand is (20/40) ÷ (10/5), which equals 1/4.



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★ Dont confuse revenue with profit

Total Revenue = P x Q

= 40 x 40 | 30 x 40

= 1600 | = 1200

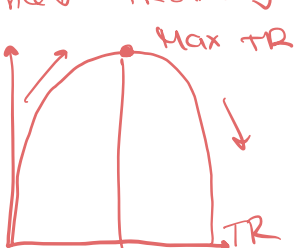
as price is affected total revenue changes

* Elasticity impacts a seller's total revenue

Price ↑ = Total Revenue ↓ } Elastic Demand (more responsive to the public)
ppl are willing to buy
Price ↓ = Total Revenue ↑
ppl wont buy

Price ↑ = Total Revenue ↑ } Inelastic Demand
Price ↓ = total Revenue ↓

Price ↑ = TR same } unit Elastic Demand
Price ↓ = TR same



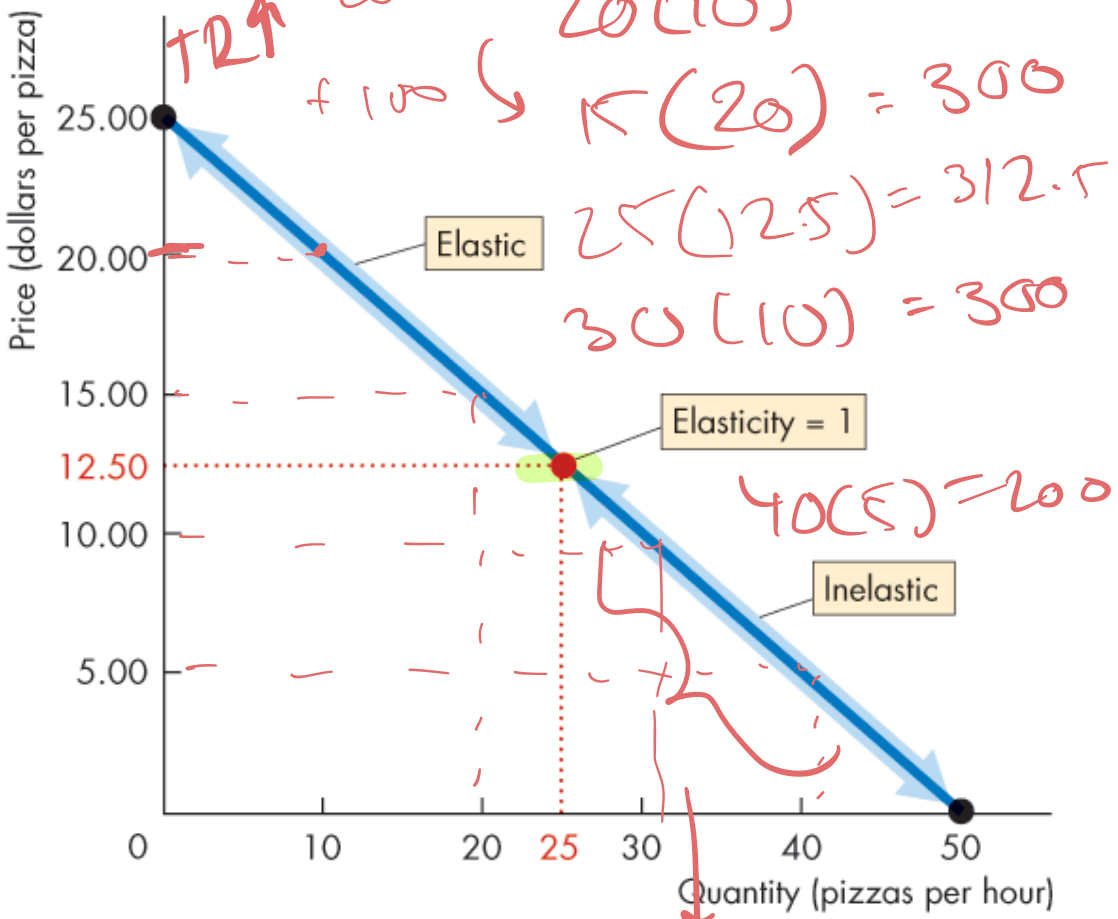
Price Elasticity of Demand

The price elasticity of demand varies along a downward–sloping linear demand curve.

At the mid-point of a linear demand curve, demand is unit elastic.

At prices above the mid-point, demand is elastic.

At prices below the mid-point, demand is inelastic.



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Price Elasticity of Demand

Total Revenue and Elasticity

The **total revenue** from the sale of a good or service equals the price of the good multiplied by the quantity sold.

When the price changes, total revenue also changes.

But a rise in price doesn't always increase total revenue.

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Price Elasticity of Demand

The change in total revenue due to a change in price depends on the elasticity of demand:

If demand is *elastic*, a 1 percent price cut increases the quantity sold by more than 1 percent, and total revenue increases. $E_d > 1$

If demand is *inelastic*, a 1 percent price cut increases the quantity sold by less than 1 percent, and total revenues decreases. $E_d < 1$

If demand is *unit elastic*, a 1 percent price cut increases the quantity sold by 1 percent, and total revenue remains unchanged. $E_d = 1$

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Price Elasticity of Demand

The **total revenue test** is a method of estimating the price elasticity of demand by observing the change in total revenue that results from a price change (when all other influences on the quantity sold remain the same).

If a price cut increases total revenue, demand is elastic. P↓ TR↑ = elastic

If a price cut decreases total revenue, demand is inelastic. P↓ TR↓ = inelastic

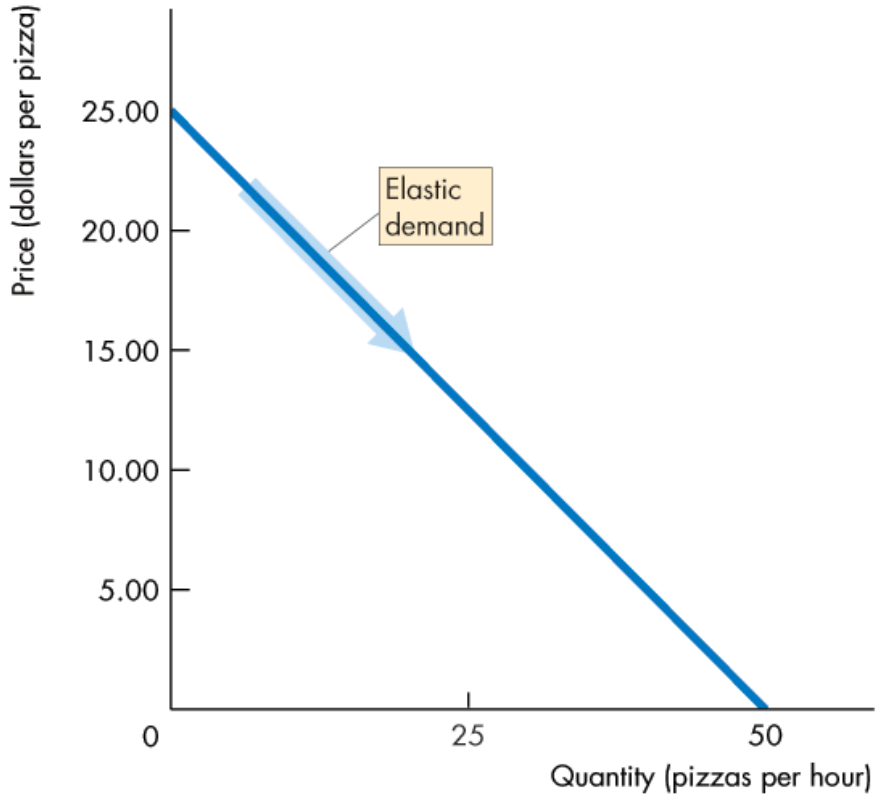
If a price cut leaves total revenue unchanged, demand is unit elastic. P↓ TR = unit elastic

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Price Elasticity of Demand

Figure 4.4 shows the relationship between elasticity of demand and the total revenue.

As the price falls from $25 toward $12.50, the quantity demanded increases and demand is elastic.

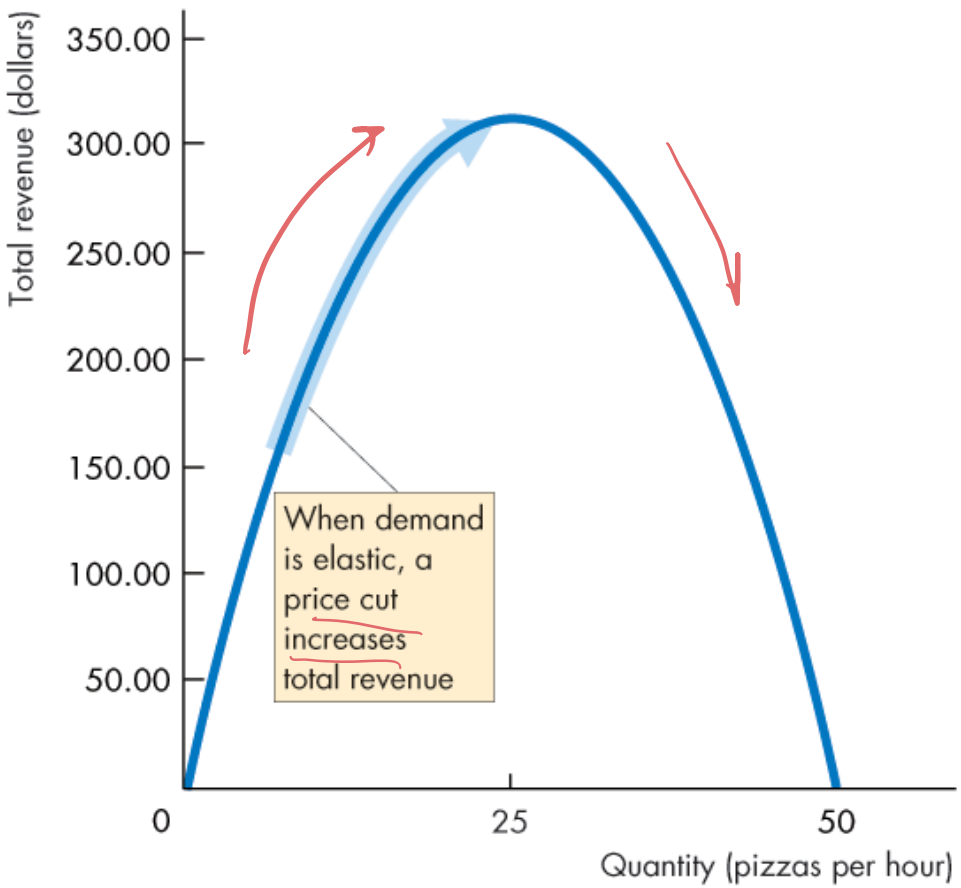


(a) Demand

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Price Elasticity of Demand

In part (b), as the quantity increases from 0 toward 25 pizzas, demand is elastic, so total revenue increases.

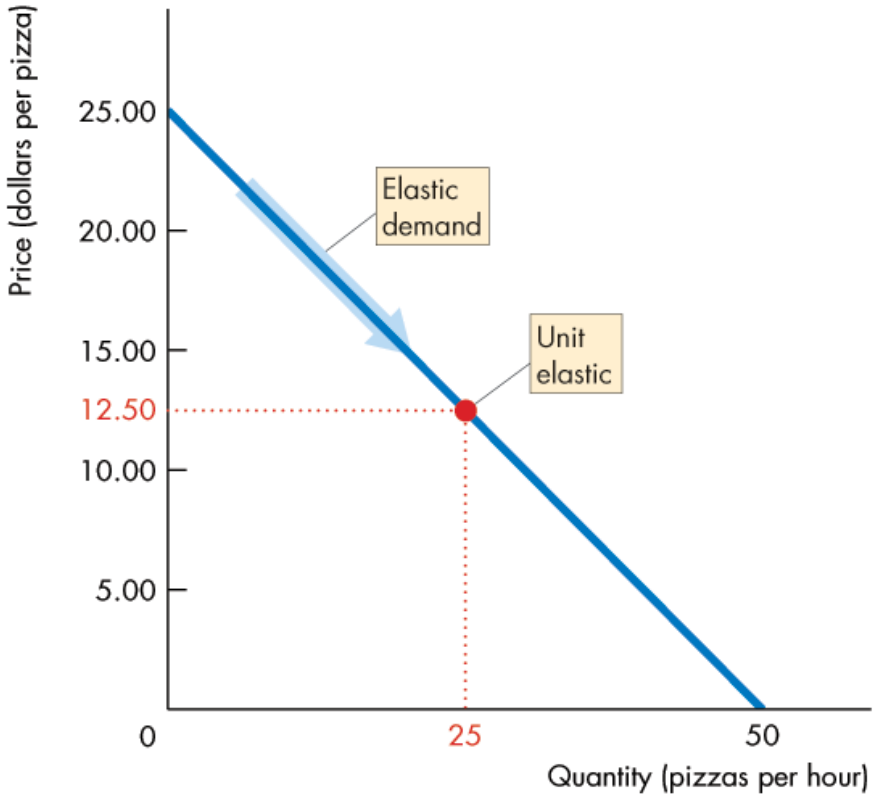


(b) Total revenue

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Price Elasticity of Demand

When the price hits $12.50 a pizza, demand is unit elastic, so total revenue stops increasing.

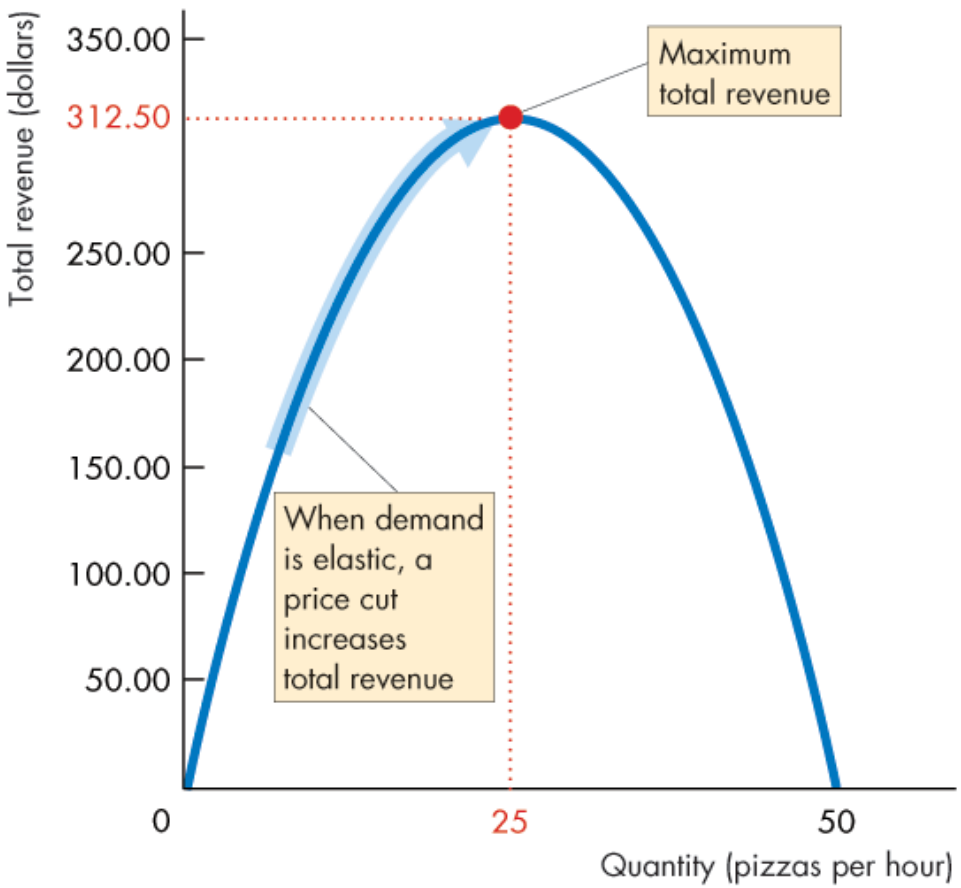


(a) Demand

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Price Elasticity of Demand

At 25 pizzas an hour, demand is unit elastic, so total revenue is at its maximum.

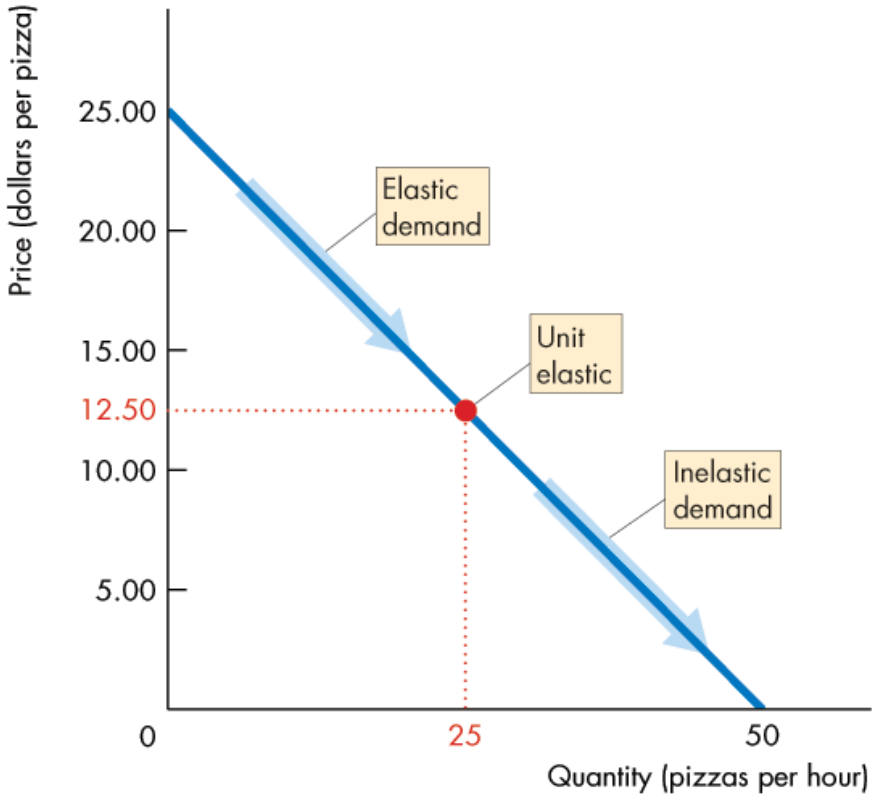


(b) Total revenue

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Price Elasticity of Demand

As the price falls below $12.50, the quantity demanded increases, and demand is inelastic.

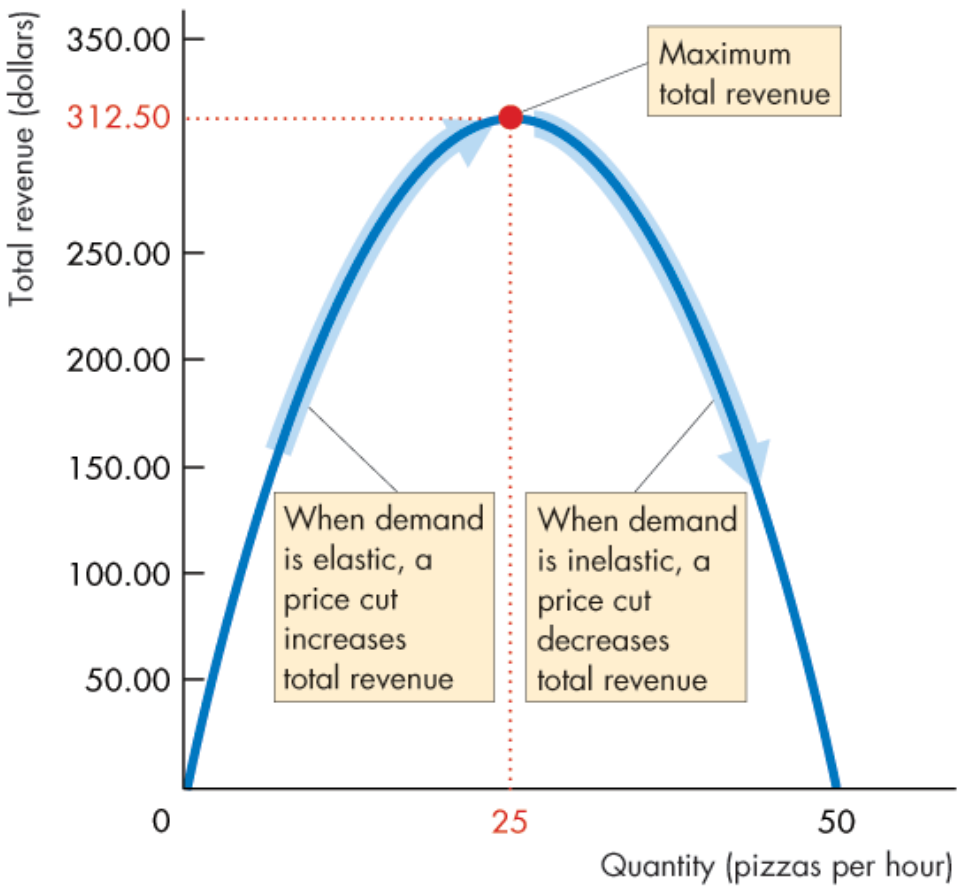


(a) Demand

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Price Elasticity of Demand

As the quantity increases from 25 toward 50 pizzas, demand is inelastic, so total revenue decreases.



(b) Total revenue

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Price Elasticity of Demand

Your Expenditure and Your Elasticity

If your demand is elastic, a 1 percent price cut increases the quantity you buy by more than 1 percent and your expenditure on the item *increases*.

If your demand is inelastic, a 1 percent price cut increases the quantity you buy by less than 1 percent and your expenditure on the item *decreases*.

If your demand is unit elastic, a 1 percent price cut increases the quantity you buy by 1 percent and your expenditure on the item *does not change*.

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Normal goods → elastic, Inferior goods → inelastic

More Elasticities of Demand

how Qd is affected relative to income

$$\frac{\%\Delta Q_d}{\%\Delta \text{income}}$$

Income Elasticity of Demand

The **income elasticity of demand** measures how the quantity demanded of a good responds to a change in income, other things remaining the same.

→ aka cetris in latin

The formula for calculating the income elasticity of demand is

$$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

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More Elasticities of Demand

$0 < E_d < 1 \rightarrow$ income inelastic

$E_d > 1 \rightarrow$ income elastic

If the income elasticity of demand is greater than 1, demand is *income elastic* and the good is a *normal good*.

If the income elasticity of demand is greater than zero but less than 1, demand is *income inelastic* and the good is a *normal good*.

If the income elasticity of demand is less than zero (negative) the good is an *inferior good*.

if income elasticity is +ve - it can be deduced that the good is a normal good while if the income elasticity is -ve - it implies that the good is inferior

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substitutes → elastic
complementary → inelastic

More Elasticities of Demand

Cross Elasticity of Demand

The **cross elasticity of demand** is a measure of the responsiveness of demand for a good to a change in the price of a *substitute* or a *complement*, other things remaining the same.

The formula for calculating the cross elasticity is:

↳ 2 products

$$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price of substitute or complement}}$$

% Δ Qd / % Δ price of substitute → % Δ Qd of coke / % Δ price of pepsi

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More Elasticities of Demand

The cross elasticity of demand for

- a *substitute* is positive.
- a *complement* is negative.

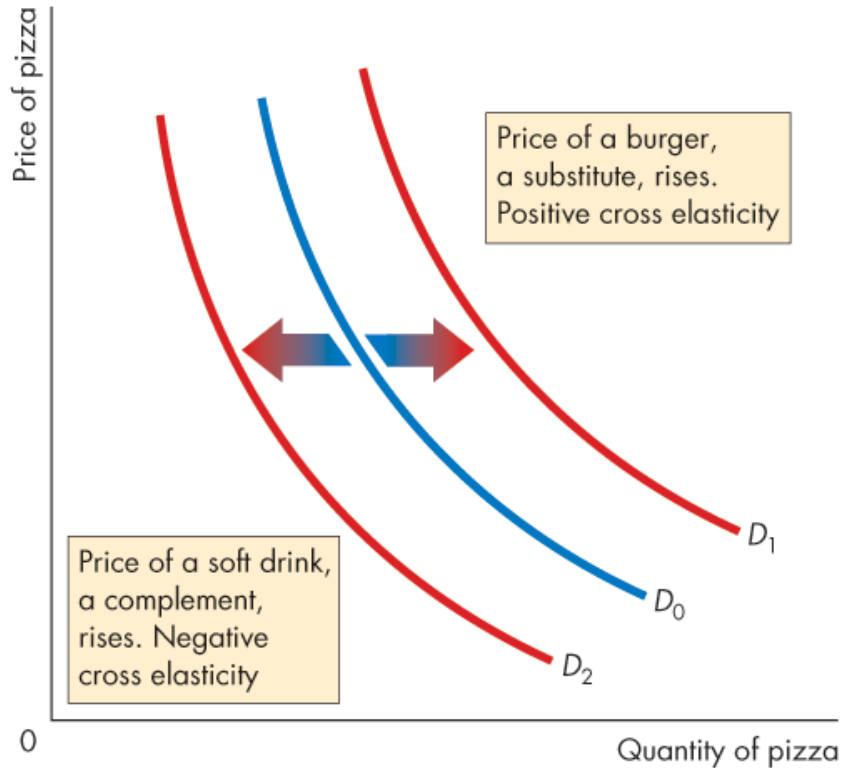
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More Elasticities of Demand

substitutes

Figure 4.5 shows the increase in the quantity demanded of pizza when the price of a burger (a substitute for pizza) rises.

The figure also shows the decrease in the quantity demanded of pizza when the price of a soft drink (a complement of pizza) rises.



↓ complementary

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Elasticity of Supply



You know that when the demand for a good increases, its equilibrium price rises and the equilibrium quantity of the good increases.



But does the price rise by a large amount and the quantity increase by a little?

Or does the price barely rise and the quantity increase by a large amount?

The answer depends on the responsiveness of the quantity supplied of a good to a change in its price.

The answer depends on the *elasticity of supply* of the good.

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Elasticity of Supply

determinant depends on timeframe

The **elasticity of supply** measures the responsiveness of the quantity supplied to a change in the price of a good, when all other influences on selling plans remain the same.

Calculating the Elasticity of Supply

The elasticity of supply is calculated by using the formula:

$$\frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

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Elasticity of Supply

Figure 4.6 on the next slide shows three examples of the elasticity of supply.

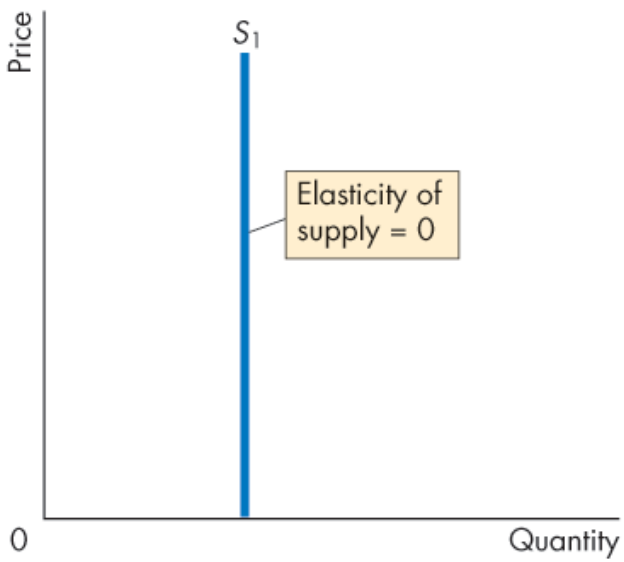
Supply is *perfectly inelastic* if the supply curve is vertical and the elasticity of supply is 0.

Supply is *unit elastic* if the supply curve is linear and passes through the origin. (Note that slope is irrelevant.)

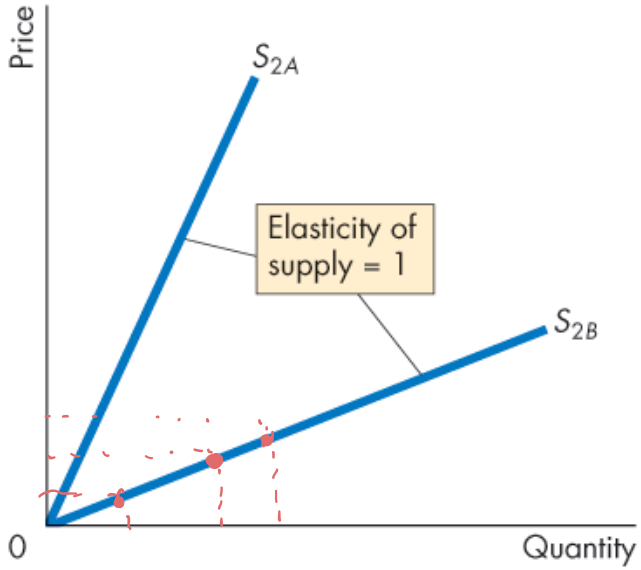
Supply is *perfectly elastic* if the supply curve is horizontal and the elasticity of supply is infinite.

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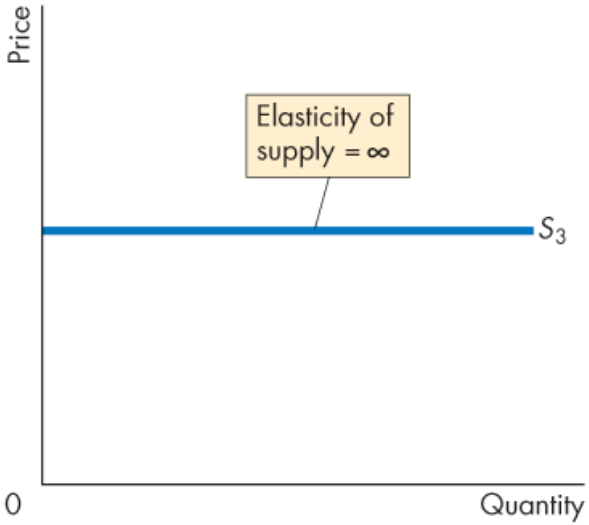
Elasticity of Supply



(a) Perfectly inelastic supply



(b) Unit elastic supply



(c) Perfectly elastic supply

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Elasticity of Supply

The Factors That Influence the Elasticity of Supply

The elasticity of supply depends on

- Resource substitution possibilities
- Time frame for supply decision

Resource Substitution Possibilities

The easier it is to substitute among the resources used to produce a good or service, the greater is elasticity of supply of the good.

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Elasticity of Supply

Time Frame for Supply Decision

As the time following a price change increases, the greater is the elasticity of supply.

Momentary supply is perfectly inelastic. The quantity supplied immediately following a price change is constant.

Short-run supply is somewhat elastic.

Long-run supply is the most elastic.

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Compact Glossary of Elasticity

Table 4.1 provides a glossary of all the elasticity measures.

TABLE 4.1 A Compact Glossary of Elasticities

Price Elasticities of Demand

A relationship is described as	When its magnitude is	Which means that
Perfectly elastic	Infinity	The smallest possible increase in price causes an infinitely large decrease in the quantity demanded*
Elastic	Less than infinity but greater than 1	The percentage decrease in the quantity demanded exceeds the percentage increase in price
Unit elastic	1	The percentage decrease in the quantity demanded equals the percentage increase in price
Inelastic	Less than 1 but greater than zero	The percentage decrease in the quantity demanded is less than the percentage increase in price
Perfectly inelastic	Zero	The quantity demanded is the same at all prices

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$0 < E_d < 1$

TABLE 4.1 A Compact Glossary of Elasticities

Income Elasticities of Demand

A relationship is described as	When its value is	Which means that
Income elastic (normal good) $E_{D\text{ of income}} > 1$	Greater than 1	The percentage increase in the quantity demanded is greater than the percentage increase in income
Income inelastic (normal good) $0 < E_d < 1$	Less than 1 but greater than zero	The percentage increase in the quantity demanded is greater than zero but less than the percentage increase in income
Negative (inferior good) $E_d = 0$	Zero	When income increases, quantity demanded decreases

TABLE 4.1 A Compact Glossary of Elasticities

Cross Elasticities of Demand

A relationship is described as	When its value is	Which means that
Close substitutes	Large	The smallest possible increase in the price of one good causes an infinitely large increase in the quantity demanded of the other good
Substitutes	Positive	If the price of one good increases, the quantity demanded of the other good also increases
Unrelated goods	Less than zero	If the price of one good increases, the quantity demanded of the other good remains the same
Complements	Negative	If the price of one good increases, the quantity demanded of the other good decreases

TABLE 4.1 A Compact Glossary of Elasticities

Elasticities of Supply

A relationship is described as	When its magnitude is	Which means that
Perfectly elastic	Infinity	The smallest possible increase in price causes an infinitely large increase in the quantity supplied*
Elastic	Less than infinity but greater than 1	The percentage increase in the quantity supplied exceeds the percentage increase in the price
Unit elastic	1	The percentage increase in the quantity supplied equals the percentage increase in the price
Inelastic	Greater than zero but less than 1	The percentage increase in the quantity supplied is less than the percentage increase in the price
Perfectly inelastic	Zero	The quantity supplied is the same at all prices

*In each description, the directions of change may be reversed. For example, in the case of a perfectly elastic demand, the smallest possible *decrease* in price causes an infinitely large *increase* in the quantity demanded.

table form calculation

P	Qd
40	70
35	75
30	80
25	85
20	90

① Find the percentages

$$\frac{\text{New - old}}{\text{Midpoint}} \times 100\%$$

$$P = \frac{75-70}{(75+70)/2} = \frac{5}{72.5} \times 100\% = 6.896 \approx 6.9\%$$

★ NEVER ROUND THE FRACTION UP